

SAKSHAM UPADHYAY

Singrauli, Madhya Pradesh 486885

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Education

VIT Bhopal University

2022 - Present

Bachelor of Technology in Electronics and Communication Engineering - 8.32/10

Bhopal

De Paul School

2021 - 2022

Intermediate 81

Singrauli

De Paul School

2019 - 2020

Secondary 77

Singrauli

Projects

1. AI PDF chatbot

- Engineered a multilingual AI PDF chatbot using LangChain and Google Gemini API, capable of processing and answering questions from 100+ page documents in 4+ languages including English, Spanish, French, and German.
- Executed a conversational pipeline integrating GoogleGenerativeAI embeddings with FAISS vector search, enabling semantic retrieval from 1000+ text chunks for accurate responses.
- Built real-time speech-to-text and text-to-speech features using SpeechRecognition and gTTS, supporting voice-based queries and audio responses for accessibility.
- Deployed a Streamlit-based UI with persistent chat history, chat export, and user feedback system, resulting in a seamless experience across 10+ end-to-end user interactions .

2. LeadMind AI

- Built Python/Streamlit lead generation platform** processing up to 50 leads per search across 5 intent categories with 0-100 scoring algorithm and 15+ keyword indicators
- Developed CRM pipeline with 4-tab interface** supporting 5 lead statuses, 3 priority levels, and 3 export formats (CSV/JSON/TXT) with real-time analytics dashboard
- Integrated Firecrawl API with automated scoring system** featuring 10+ high-intent keywords, 6 buying signals, and 5 urgency indicators for lead qualification
- Engineered responsive UI with custom CSS featuring 25+ styled components**, 4-color priority system, gradient backgrounds, and automated progress tracking with 7-day recent activity filters and conversion rate calculations

3. Deep Fake Audio Detection

- Architected a CRNN model achieving 73%+ accuracy** in detecting audio deepfakes with 3 convolutional layers and a 2-layer bidirectional LSTM, trained on 1,000+ real/fake audio samples.
- Optimized audio preprocessing** by standardizing variable-length inputs to fixed 200-frame spectrograms (16kHz SR, 1024 n_fft), reducing inference latency by 30% through GPU-accelerated batch processing (16 samples/batch).
- Reduced overfitting by 15%** via dropout (0.3) and adaptive learning rate scheduling (patience=2), while maintaining 90%+ validation accuracy across 10 training epochs.

Technical Skills

Languages: Java, Python, HTML, SQL

Developer Tools: VS Code

Libraries/Frameworks: NumPy, Pandas, Scikit-Learn, TensorFlow, Keras, Tailwind CSS

AI/ML and NLP Tools: Machine Learning, Generative AI, Natural Language Processing (NLP), LangChain, LanGraph, CrewAI

Databases: MySQL

Certifications

- HTML, CSS, and Javascript for Web Developers
- Java certification by SkillVertex
- GenAI certification by IBM